Unit-II: Correlation and Regression

1. Correlation

A correlation is a statistical measure of the relationship between two variables. The measure is best used in variables that demonstrate a linear relationship between each other. The fit of the data can be visually represented in a scatterplot. Using a scatterplot, we can generally assess the relationship between the variables and determine whether they are correlated or not.

Covariance: Covariance is a measure of the relationship between two random variables.

In Individual Series: $\mathrm{cov}(\mathrm{x},\mathrm{y}) = \frac{1}{\mathrm{n}}\sum(\mathrm{x}-\overline{\mathrm{x}})\cdot(\mathrm{y}-\overline{\mathrm{y}}) = \frac{1}{\mathrm{n}}\sum \mathrm{xy} - \overline{\mathrm{x}}\,\overline{\mathrm{y}}$

In Continuous Series*:* If $u = \frac{x-A}{h}$ *and* $v = \frac{y-B}{k}$ then $cov(u,v) = \frac{1}{n}\sum(u-\overline{u})\cdot(v-\overline{v})$ *or* $cov(u,v) = \frac{1}{n}\sum uv - \overline{u}\,\overline{v}$

Karl Pearson's Coefficient of Correlation: Denoted by $r(x,y)$ and defined as

$$1.\, r(x,y) = \frac{cov(x,y)}{\sigma_x\sigma_y} = \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$$

$$2.\, If\ u = \frac{x-A}{h}\ and\ v = \frac{y-B}{k}\ then\ r(u,v) = \frac{cov(u,v)}{\sigma_u\sigma_v}$$

$$= \frac{\sum uv - n\overline{uv}}{\sqrt{(\sum u^2 - n(\overline{u})^2)(\sum v^2 - n(\overline{v})^2)}}$$

Range of $r(x,y)\ is\ -1 \le r \le 1.$

Positive Correlation: As one set of values increases while the other set increases then it is called a positive correlation. $r(x,y) > 0.$

Negative Correlation: As one set of values increases while the other set decreases then it is called a negative correlation. $r(x,y) < 0.$

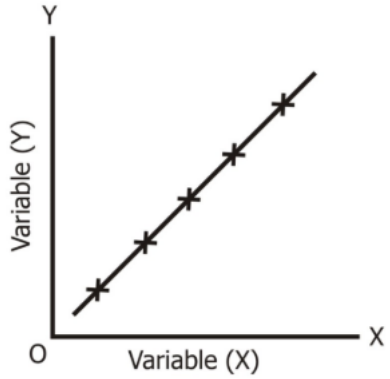
No/Zero Correlation: If the change in values of one set does not affect the values of the other, then the variables are said to have "no correlation" or "zero correlation." $r(x,y) = 0.$

Scatter Diagram

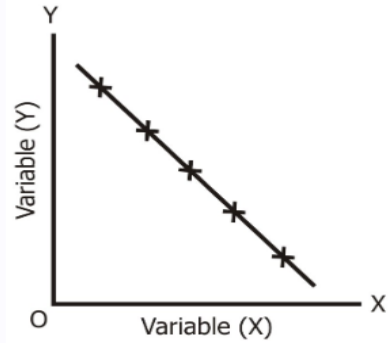
A scatter diagram is a tool for analyzing relationships between two variables. One variable is plotted on the horizontal axis and the other is plotted on the vertical axis. The pattern of their intersecting points can graphically show relationship patterns. Most often a scatter diagram is used to prove or disprove cause-and-effect relationships. While the diagram shows relationships, it does not by itself prove that one variable causes the other. In addition to showing possible cause and-effect relationships, a scatter diagram can show that two variables are from a common cause that is unknown or that one variable can be used as a surrogate for the other.

Use a scatter diagram to examine theories about cause-and-effect relationships and to search for root causes of an identified problem. Use a scatter diagram to design a control system to ensure that gains from quality improvement efforts are maintained.

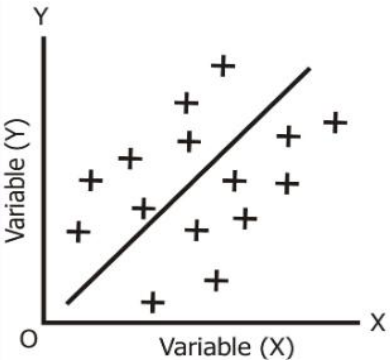
Perfect Positive Correlation: If the points of the scatter diagram fall on a straight line and have a positive (upward) slope, then the correlation is said to be perfectly positive; i.e., $\mathrm{r} = +1.$



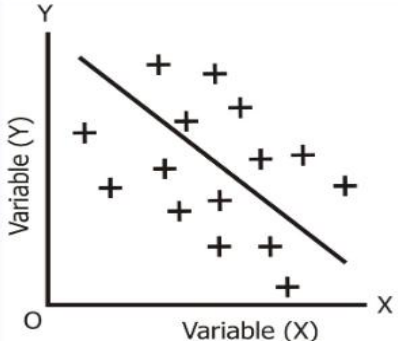
Perfect Negative Correlation: If the points of the scatter diagram fall on a straight line and have a negative (downward) slope, then the correlation is said to be perfectly negative; i.e., $r = -1$.



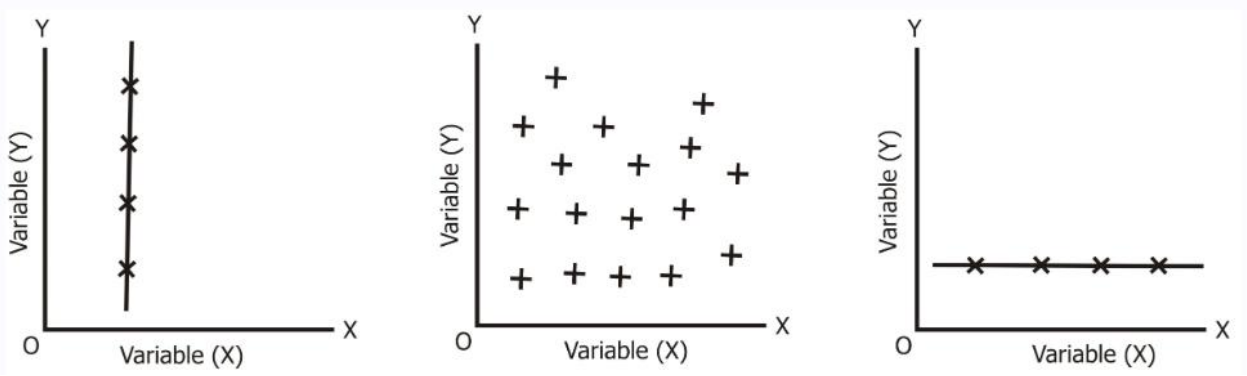
Weak Positive Correlation: When the points of the scatter diagram cluster around a straight line (upward slope from left to right), then the correlation is said to be positive.



Weak Negative Correlation: When the points of the scatter diagram cluster around a straight line (downward/negative slope), then the correlation is said to be negative.



No Correlation: When the points of the scatter diagram are scattered in a haphazard manner, then there is zero or no correlation.



Examples:

1. Find the co-efficient of correlation for the data n=10, $\overline{x} = 5.5, \overline{y} = 4, \sum x^2 = 385,$ $\sum y^2 = 192, \sum(x+y)^2 = 947.$

   Solution: use r(x, y) $= \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

   First find $\sum xy$

   Consider$\sum(x+y)^2 = 947$

   $\therefore \sum(x^2 + 2xy + y^2) = 947$

   $\therefore \sum x^2 + 2\sum xy + \sum y^2 = 947$

   $\therefore 2\sum xy = 947 - \sum x^2 - \sum y^2$

   $\therefore 2\sum xy = 947 - 385 - 192 = 370$

   $\therefore \sum xy = 185$

   $\therefore r(x,y) = \frac{185 - 10(5.5)(4)}{\sqrt{(385 - 10(5.5)^2)(192 - 10(4)^2)}}$

   $\therefore r(x,y) = -0.681.$

2. Find the co-efficient of correlation between x and y for the following data,

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

   Solution: Consider r(x, y) $= \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

x	y	x^2	y^2	xy
1	1	1	1	1
3	2	9	4	6
4	4	16	16	16
6	4	36	16	24
8	5	64	25	40
9	7	81	49	63

11	8	121	64	88
14	9	196	81	126
56	40	524	256	364

Let $n = 8, \overline{x} = \frac{\sum x}{n} = \frac{56}{8} = 7$and $\overline{y} = \frac{\sum y}{n} = \frac{40}{8} = 5$

Use formula r(x, y) $= \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

$\therefore r(x, y) = \frac{364-8(7)(5)}{\sqrt{(524-8(7)^2)(256-8(5)^2)}} = \frac{84}{\sqrt{132}\sqrt{56}} = 0.977.$

3. Obtain the correlation between population density and death rate.

Population density	200	400	500	700	300
Death rate	12	18	16	21	10

Solution: Consider r(x, y) $= \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

x	y	x^2	y^2	xy
200	12	40000	144	2400
400	18	160000	324	7200
500	16	250000	256	8000
700	21	490000	441	14700
300	10	90000	100	3000
2100	**77**	**1030000**	**1265**	**35300**

Let n=5, $\overline{x} = \frac{\sum x}{n} = \frac{2100}{5} = 420$and $\overline{y} = \frac{\sum y}{n} = \frac{77}{5} = 15.4$

$\therefore r(x, y) = \frac{35300 - 5(420)(15.4)}{\sqrt{(1030000 - 5(420)^2)(1265 - 5(15.4)^2)}}$

$\therefore r(x, y) = \frac{2960}{\sqrt{148000}\sqrt{79.2}} = 0.8645$

4. Calculate the coefficient of correlation between the import and export values.

Export	10	11	14	14	20	22	16	12	15	13
Import	12	14	15	16	21	26	21	15	16	14

Solution: Consider r(x,y) $= \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

x	y	x^2	y^2	xy
10	12	100	144	120
11	14	121	196	154
14	15	196	225	210
14	16	196	256	224
20	21	400	441	420

22	26	484	676	572
16	21	256	441	336
12	15	144	225	180
15	16	225	256	240
13	14	169	196	182
147	**170**	**2291**	**3056**	**2638**

Let n=10, $\overline{x} = \frac{\sum x}{n} = \frac{147}{10} = 14.7$and $\overline{y} = \frac{\sum y}{n} = \frac{170}{10} = 17$

$$r(x, y) = \frac{2638 - 10(14.7)(17)}{\sqrt{(2291 - 10(14.7)^2)(3056 - 10(17)^2)}} = \frac{139}{\sqrt{130.1}\sqrt{166}} = 0.9458.$$

5. Find correlation coefficient between supply and price of from following data.

supply	152	158	169	182	160	166	182
Price	198	178	167	152	180	170	162

Solution: Consider $r(x, y) = \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}}$

x	y	x^2	y^2	xy
152	198	23104	39204	30096
158	178	24964	31684	28124
169	167	28561	27889	28223
182	152	33124	23104	27664
160	180	25600	32400	28800
166	170	27556	28900	28220
182	162	33124	26244	29484
1169	**1207**	**196033**	**209425**	**200611**

Let n=7, $\overline{x} = \frac{\sum x}{n} = \frac{1169}{7} = 167$and $\overline{y} = \frac{\sum y}{n} = \frac{1207}{7} = 172.42$

$$r(x, y) = \frac{\sum xy - n\overline{xy}}{\sqrt{(\sum x^2 - n(\overline{x})^2)(\sum y^2 - n(\overline{y})^2)}} = \frac{-958}{\sqrt{810}\sqrt{1303.714}} = -0.9322.$$

2. Regression

Linear regression is a form of statistical analysis that attempts to identify a relationship between an independent variable and one or more dependent variables.

Generally linear regression is aimed at causal relationships, where a change in the independent variable causes a predictable change in the dependent variable.

Line of Regression of Y on X:

$y - \overline{y} = \frac{cov(x,y)}{\sigma_x^2}(x - \overline{x})$ or $y - \overline{y} = \frac{r(x,y)\sigma_y}{\sigma_x}(x - \overline{x})$ or $y - \overline{y} = b_{yx}(x - \overline{x})$

Where $b_{yx} = \frac{cov(x,y)}{\sigma_x^2}$ or $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ is coefficient of regression line of y on x.

Line of Regression of X on Y:

$x - \overline{x} = \frac{cov(x,y)}{\sigma_y^2}(y - \overline{y})$ or $x - \overline{x} = \frac{r(x,y)\sigma_x}{\sigma_y}(y - \overline{y})$ or $x - \overline{x} = b_{xy}(y - \overline{y})$

Where $b_{xy} = \frac{cov(x,y)}{\sigma_y^2}$ or $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ is coefficient of regression line of x on y.

Properties:

- $r = \sqrt{b_{xy} \cdot b_{yx}}$ or $r^2 = b_{xy} \cdot b_{yx}$
- Angle between two regression lines: $tan\,\theta = \frac{1-r^2}{|r|} \cdot \frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$.

Examples:

1. Find the regression lines and estimate y for x= 14.5 and x for y=29.5.

x	10	14	19	26	30	34	39
Y	12	16	18	26	29	35	38

Solution: Let n=7, first find mean of x and y.

x	y	x^2	y^2	xy
10	12	100	144	120
14	16	196	256	224
19	18	361	324	342
26	26	676	676	676
30	29	900	841	870
34	35	1156	1225	1190
39	38	1521	1444	1482
172	174	4910	4910	4904

$\overline{x} = \frac{\sum x}{n} = \frac{172}{7} = 24.57$ and $\overline{y} = \frac{\sum y}{n} = \frac{174}{7} = 24.85$

$$\sigma_x = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = \sqrt{\frac{4910}{7} - (24.57)^2} = 9.883$$

$$\sigma_y = \sqrt{\frac{1}{n}\sum y^2 - (\overline{y})^2} = \sqrt{\frac{4910}{7} - (24.85)^2} = 9.14$$

$$cov(x,y) = \frac{1}{n}\sum xy - \overline{xy} = \frac{4904}{7} - (24.57)(24.85) = 89.795$$

$$r(x,y) = \frac{cov(x,y)}{\sigma_x\sigma_y} = \frac{89.795}{9.883 * 9.14} = 0.9941$$

$b_{yx} = r\frac{\sigma_y}{\sigma_x} = 0.9941\frac{9.14}{9.883} = 0.9193$ and $b_{xy} = r\frac{\sigma_x}{\sigma_y} = 0.9941\frac{9.883}{9.14} = 1.0749$

Regression line of y on x is given by equation: $y - \overline{y} = b_{yx}(x - \overline{x})$

$\therefore y - 24.85 = 0.9193(x - 24.57) = 0.9193x - 22.58965$

$\therefore y = 0.9193x - 22.58965 + 24.85$

$\therefore y = 0.9193x + 2.26745$

To estimate y at x=14.5

$\therefore y = 0.9193(14.5) + 2.26745 = 15.5977$

Regression line of x on y is given by equation:$x - \overline{x} = b_{xy}(y - \overline{y})$

$\therefore x - 24.57 = 1.0749(y - 24.85) = 1.0749y - 26.71894$

$\therefore x = 1.0749y - 26.71894 + 24.57$

$\therefore x = 1.0749y - 2.14751$

To estimate x for y=29.5. So put y=29.5 in above equation

$\therefore x = 1.0749(29.5) - 2.14751 = 29.56176.$

2. Obtain line of regression for following data and estimate y for x=4.5.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Solution: Let n=9, first find mean of x and y.

x	y	x^2	y^2	xy
1	9	1	81	9
2	8	4	64	16
3	10	9	100	30
4	12	16	144	48
5	11	25	121	55
6	13	36	169	78
7	14	49	196	98
8	16	64	256	128
9	15	81	225	135
45	108	285	1356	597

$\overline{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$ and $\overline{y} = \frac{\sum y}{n} = \frac{108}{9} = 12$

$$\sigma_x = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = \sqrt{\frac{285}{9} - (5)^2} = 2.5819$$

$$\sigma_y = \sqrt{\frac{1}{n}\sum y^2 - (\overline{y})^2} = \sqrt{\frac{1356}{9} - (12)^2} = 2.5819$$

$$cov(x,y) = \frac{1}{n}\sum xy - \overline{xy} = \frac{597}{9} - (5)(12) = 6.3333$$

$$r(x,y) = \frac{cov(x,y)}{\sigma_x \sigma_y} = \frac{6.3333}{2.5819 * 2.5819} = 0.94274$$

$b_{yx} = r\frac{\sigma_y}{\sigma_x} = 0.94274\frac{2.5819}{2.5819} = 0.94274$ and $b_{xy} = r\frac{\sigma_x}{\sigma_y} = 0.94274\frac{2.5819}{2.5819} = 0.94274$

Regression line of y on x is given by equation: $y - \overline{y} = b_{yx}(x - \overline{x})$

$\therefore y - 12 = 0.94274(x - 5) = 0.94274x - 4.7137$

$\therefore y = 0.94274x - 4.7137 + 12$

$\therefore y = 0.94274x + 7.28625$

To estimate y for x=4.5. So put x=4.5 in above equation

$\therefore y = 0.94274(4.5) + 7.28625 = 11.52863.$

3. Two lines of regression are$5y - 8x + 17 = 0$, 2y-5x + 14 $= 0$. If$\sigma_y^2 = 16$.
   Find: (i) The mean value of x and y (ii) Correlation co-efficient $r(x, y)$ (iii)σ_x.

   Solution: (i) To find the mean value of x and y.

   Since both the regression lines pass through the point $(\overline{x}, \overline{y})$ i. e. put$x = \overline{x}$ and y $= \overline{y}$.

   $5\overline{y} - 8\overline{x} = -17$ (1)

   $2\overline{y}$-$5\overline{x} = -14$ (2)

   Multiply eq^n (1) by 2 – Multiply eq^n (2) by 5

   $(10\overline{y} - 16\overline{x} + 34) - (10\overline{y}\text{-}25\overline{x} + 70) = 0 - 0$

   $\therefore 10\overline{y} - 16\overline{x} + 34 - 10\overline{y} + 25\overline{x} - 70 = 0$

   $\therefore 9\overline{x} - 36 = 0$

   $\therefore \overline{x} = 4$ Put in eq^n (1) and we get $\overline{y} = 3$

   (ii) To find Correlation co-efficient r(x, y).

   Let $5y - 8x + 17 = 0$ be the line of regression of y on x.

   $5y - 8x + 17 = 0 \therefore 5y = 8x - 17 \therefore y = \frac{8}{5}x - \frac{17}{5}$ $\therefore y = 1.6x - 3.4$ (3)

   Let 2y-5x + 14 $= 0$ be the line of regression of x on y.

   2y-5x + 14 $= 0 \therefore$ 5x = 2y + 14 $\therefore x = \frac{2}{5}y + \frac{14}{5}$ $\therefore x = 0.4y + 2.8$ (4)

   Let $b_{yx} = 1.6$Regression Coefficient of y on x.

   Let $b_{xy} = 0.4$Regression Coefficient of x on y.

   Correlation Coefficient between x and y is

   $r = \sqrt{b_{xy} \cdot b_{yx}} = \sqrt{0.4 * 1.6} = \sqrt{0.64} = 0.8$

   (iii) To findσ_x. Given that $\sigma_y^2 = 16 \Rightarrow \sigma_y = 4$

   Consider $b_{yx} = r\frac{\sigma_y}{\sigma_x}$

   $\therefore \sigma_x = r\frac{\sigma_y}{b_{yx}} = 0.8\frac{4}{1.6} = 2.$

4. The two-regression equation of the variables x and y are $x = 19.13 - 0.87y$and $y = 11.64 - 0.50x$Find (i) $\overline{x}$ and $\overline{y}$ (ii) The Correlation Coefficient between x and y.

   Solution: (i) To find the mean value of x and y.

   Since both the regression lines pass through the point $(\overline{x}, \overline{y})$ that is put$x = \overline{x}$ and y $= \overline{y}$.

   $\overline{x} = 19.13 - 0.87\overline{y}$ $\therefore \overline{x} + 0.87\overline{y} = 19.13$ (1)

   $\overline{y} = 11.64 - 0.5\overline{x}$ $\therefore 0.5\overline{x} + \overline{y} = 11.64$ (2)

   Multiply eq^n (1) by 0.5 – Multiply eq^n (2) by 1

   $(0.5\overline{x} + 0.435\overline{y} = 9.565) - (0.5\overline{x} + \overline{y} = 11.64)$

   $\therefore -0.565\overline{y} = -2.075$

   $\therefore \overline{y} = 3.673$ Put in eq^n (1) $\therefore \overline{x} = 15.935$

   (ii) To find Correlation co-efficient r(x, y).

Let $y = -0.50x + 11.64$ be the line of regression of y on x.
Let $b_{yx} = -0.5$ Regression Coefficient of y on x.
Let $x = -0.87y + 19.13$ be the line of regression of x on y.
Let $b_{xy} = -0.87$ Regression Coefficient of x on y.
Correlation Coefficient between x and y is

$$r = \sqrt{b_{xy} \cdot b_{yx}} = \sqrt{-0.5 * -0.87} = \sqrt{0.64} = 0.6595$$

5. The regression lines are $5x - 6y + 90 = 0$ and 15x-8y-130 $= 0$, the variance of x is 9. Find (i) the mean values of x and y. (ii) The correlation x and y. (iii) The S.D of y.

Solution: (i) To find the mean value of x and y.
Since both the regression lines pass through the point $(\overline{x}, \overline{y})$ i. e. put$x = \overline{x}$ and y $= \overline{y}$.
$5\overline{x} - 6\overline{y} = -90$ (1)
$15\overline{x} - 8\overline{y} = 130$ (2)
Multiply eq^n (1) by 3 – Multiply eq^n (2) by 1
$(15\overline{x} - 18\overline{y} = -270) - (15\overline{x} - 8\overline{y} = 130)$
$\therefore$ -10$\overline{y} = -400$
$\therefore \overline{y} = 40$ Put in eq^n (1) $\therefore \overline{x} = 48.3333$
(ii) To find Correlation co-efficient $r(x, y)$.
Let $5x - 6y + 90 = 0$ be the line of regression of y on x.
$\therefore 6y = 5x + 90$ $\therefore y = \frac{5}{6}x + \frac{90}{6} \therefore y = 0.833x + 15$ (3)
Let $b_{yx} = 0.8333$Regression Coefficient of y on x.
Let 15x-8y-130 $= 0$ be the line of regression of x on y.
$\therefore$ 15x $= 8y + 130 \therefore x = \frac{8}{15}y + \frac{130}{15}$ $\therefore x = 0.533y + 8.6667$ (4)
Let $b_{xy} = 0.533$Regression Coefficient of x on y.
Correlation Coefficient between x and y is

$$r = \sqrt{b_{xy} \cdot b_{yx}} = \sqrt{0.8333 * 0.533} = \sqrt{0.4444} = 0.6666$$

(iii) To findσ_y. Given that variance of x is 9 i. e $\sigma_x^2 = 9 \Rightarrow \sigma_x = 3$
Consider $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ $\therefore \sigma_y = \frac{b_{yx}\sigma_x}{r} = \frac{0.83333*3}{0.6666} = 3.75.$

3. Curve Fitting

Least Square Method

The method of least squares is the most systematic procedure to fit a unique curve (or line) through the given points.

Fitting of Straight Line

A) To fit the straight line $y = ax + b$ by using their normal equations
$\sum y = a\sum x + bn$ and
$\sum xy = a\sum x^2 + b\sum x$
B) Reduce the calculations by using $x = X + h$ or $X = x - h$ where h is generally middle value of observed values of x, then use their normal equations

$\sum y = a\sum X + nb$ and
$\sum Xy = a\sum X^2 + b\sum X$
Using above equations find a, b and put it in the given equation $y = ax + b$.
C) Extra formula for finding a and b.

$$a = \frac{\sum x \sum y - n\sum xy}{(\sum x)^2 - n\sum x^2} \text{ and } b = \frac{\sum y}{n} - a\frac{\sum x}{n} = \frac{\sum y - a\sum x}{n}.$$

Fitting of Parabola

Fitting of Parabola $y = ax^2 + bx + c$

A) To fit the parabola of the form $y = ax^2 + bx + c$ by using their normal equations
$\sum y = a\sum x^2 + b\sum x + nc$
$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$
$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2$
B) Reduce the calculations by using $x = X + h$ or $X = x - h$ where h is generally middle value of observed values of x.
$\sum y = a\sum X^2 + b\sum X + nc$
$\sum Xy = a\sum X^3 + b\sum X^2 + c\sum X$
$\sum X^2y = a\sum X^4 + b\sum X^3 + c\sum X^2$
Using above equations find a, b, c and put it in the given equation $y = ax^2 + bx + c$.

Fitting of Parabola $x = ay^b$

To fit the parabola of the form $x = ay^b$ taking logarithms on both sides
$logx = log(ay^b)$
$logx = log(a) + log(y^b)$
$logx = loga + b\,logy$
$logx = b\,logy + log\,a$
$X = bY + c$ where $Y = logy, X = logx$ and $c = loga$.
Use their normal equations.
$\sum X = b\sum Y + cn$ and $\sum XY = b\sum Y^2 + c\sum Y$
Using above equation find values of a, b and put it in the given equation $x = ay^b$.

Fitting of Parabola $y = ax^b$

To fit the parabola of the form $y = ax^b$ taking logarithms on both sides
$logy = log(ax^b)$
$logy = log(a) + log(x^b)$
$logy = loga + b\,logx$
$logy = b\,logx + log\,a$
$Y = bX + c$ where $Y = logy, X = logx$ and $c = loga$
Use their normal equations.
$\sum Y = b\sum X + cn$ and $\sum XY = b\sum X^2 + c\sum X$
Using above equation find values of a, b and put it in the given equation $x = ay^b$.

Fitting of Parabola $y = a\,b^x$

To fit the parabola of the form $y = a\,b^x$ taking logarithms on both sides

$logy = log(ab^x)$
$logy = log(a) + log(b^x)$
$logy = \ loga \ + x \ logb$
$logy \ = \ x \ logb \ + \ log \ a$
$Y = Bx + C$ where $B = logb, C = loga$ and $Y = logy$.
Use their normal equations.
$\sum Y = B \sum x + nC$ and
$\sum xY = B \sum x^2 + C \sum x$
Using above equation find values of a, b and put it in the given equation y = $a \ b^x$.

Fitting of Parabola $\mathbf{ax^2 + by^2 = x}$
To fit the parabola of the form $ax^2 + by^2 \ = x$
Use their normal equations.
$a \sum x^4 + b \sum x^2y^2 - \sum x^3 = 0$ and
$a \sum x^2y^2 + b \sum y^4 - \sum x \ y^2 = 0$
Using above equation find a, b and put it in the given equation $ax^2 + by^2 \ = x$.

Table for the equation of curve and their normal equations.

Sr. No.	Equation of curve	Normal equations
1	Fitting Straight Line $y = ax + b$	$\sum y = a \sum x + bn$ and $\sum xy = a \sum x^2 + b \sum x$
2	Fitting the Parabola $y = ax^2 + bx + c$	$\sum y = a \sum x^2 + b \sum x + nc$ $\sum$ xy =a $\sum$ x^3+b $\sum$ x^2 +c $\sum$ x $\sum x^2y = a \sum x^4 + b \sum x^3 + c \sum x^2$
3	Fitting the Parabola $x = ay^b$	$\sum X = b \sum Y + cn$ and $\sum XY = b \sum Y^2 + c \sum Y$ where $Y = logy, X = \ logx \ \& c = loga$
4	Fitting the Parabola $y = ax^b$	$\sum Y = b \sum X + cn$ and $\sum XY = b \sum X^2 + c \sum X$ Where $Y = logy, X = \ logx$ & $c = loga$
5	Fitting the Parabola $y = a \ b^x$	$\sum Y = B \sum x + nC$ and $\sum xY = B \sum x^2 + C \sum x$ where $B = logb, C = loga$ & $Y = logy$
6	Fitting the Parabola $ax^2 + by^2 = x$	$a \sum x^4 + b \sum x^2y^2 - \sum x^3 = 0$ and $a \sum x^2y^2 + b \sum y^4 - \sum x \ y^2 = 0$

Examples:

Unit-II: Correlation and Regression

1. Fit a straight line of the form $y = mx + c$ to the following data, by using the method of least squares.

x	0	1	2	3	4	5	6	7
y	-5	-3	-1	1	3	5	7	9

Solution: To fit the straight line $y = mx + c$ by using their normal equations, let n=8

$\sum y = m\sum x + cn$ (1)

$\sum xy = m\sum x^2 + c\sum x$ (2)

x	y	xy	x^2
0	-5	0	0
1	-3	-3	1
2	-1	-2	4
3	1	3	9
4	3	12	16
5	5	25	25
6	7	42	36
7	9	63	49
28	**16**	**140**	**140**

From equation (1), (2) and table, we have

$16 = 28m + 8c \ \ or \ \ 28m + 8c = 16$

$140 = 140m + 28c \ \ or \ \ 140m + 28c = 140$

Solving these two equations, we get, $m = 2 \ and \ c = -5$ put these values in given equation, therefore $y = mx + c = 2x - 5$.

2. For the following values of x and y fit a linear curve of the type $y = ax + b$

x	1.0	3.0	5.0	7.0	9.0
y	1.5	2.8	4.0	4.7	6.0

Solution: To fit the straight line $y = ax + b$ by using their normal equations, let n=5

$\sum y = a\sum x + bn$ …1

$\sum xy = a\sum x^2 + b\sum x$ …2

x	y	xy	x^2
1	2	2	1
3	3	9	9
5	4	20	25
7	5	35	49
9	6	54	81
25	**20**	**120**	**165**

From equation (1), (2) and table, we have

$\sum y = a\sum x + bn \implies 25a + 5b = 20$

$\sum xy = a\sum x^2 + b\sum x \implies 165a + 25b = 120$
Solving these two equations, we get, $a = 0.5\ and\ b = 1.5$ put these values in given equation, therefore $y = ax + b = 0.5x + 1.5$.

3. Find the best values of a, b, c assuming that the values of x and y are connected by the relation $y = ax^2 + bx + c$

x	1	2	3	4	5
y	3.38	8.25	16.6	28.5	44

Solution: To fit parabola $y = ax^2 + bx + c$ by using their normal equations & let $n = 5$
$\sum y = a\sum x^2 + b\sum x + nc$ ...1
$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$ ...2
$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2$ ...3

x	y	x^2	x^3	x^4	xy	x^2y
1	*3.38*	*1*	*1*	*1*	*3.38*	*3.38*
2	*8.25*	*4*	*8*	*16*	*16.5*	*33*
3	*16.6*	*9*	*27*	*81*	*49.8*	*149.4*
4	*28.5*	*16*	*64*	*256*	*114*	*456*
5	*44*	*25*	*125*	*625*	*220*	*1100*
15	***100.73***	***55***	***225***	***979***	***403.68***	***1741.78***

From above equations and table, we have
$\sum y = a\sum x^2 + b\sum x + nc \implies 55a + 15b + 5c = 100.73$
$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x \implies 225a + 55b + 15c = 403.68$
$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2 \implies 979a + 225b + 55c = 1741.8$
Solving these equations, we get, $a = 1.773,\ b = -0.5\ and\ c = 2.11$ put these values in given equation, therefore $y = 1.77x^2 - 0.5x + 2.11$.

4. Fit a parabola of the form $y = ax^2 + bx + c$ to the following data, by using the method of least squares.

x	1	2	3	4	5	6	7
y	-5	-2	5	16	31	50	73

Solution: To fit parabola $y = ax^2 + bx + c$ by using their normal equations & let $n = 7$
$\sum y = a\sum x^2 + b\sum x + nc$ ...1
$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$ ...2
$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2$ ...3

x	y	x^2	x^3	x^4	xy	x^2y
1	*-5*	*1*	*1*	*1*	*-5*	*-5*
2	*-2*	*4*	*8*	*16*	*-4*	*-8*
3	*5*	*9*	*27*	*81*	*15*	*45*
4	*16*	*16*	*64*	*256*	*64*	*256*
5	*31*	*25*	*125*	*625*	*155*	*775*
6	*50*	*36*	*216*	*1296*	*300*	*1800*

7	*73*	*49*	*343*	*2401*	*511*	*3577*
28	***168***	***140***	***784***	***4676***	***1036***	***6440***

From above equations and table, we have

$\sum y = a\sum x^2 + b\sum x + nc \implies 140a + 28b + 7c = 168$

$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x \implies 784a + 140b + 28c = 1036$

$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2 \implies 4676a + 784b + 140c = 6440$

Solving these equations,

we get, $a = 2,\ b = -3\ and\ c = -4$ and put these values in given equation,

$\therefore y = 2x^2 - 3x - 4.$

5. A simply supported beam carries a concentrated load p (kg) its middle point. Corresponding to various of p, the maximum deflection y cm is calculated as:

p	100	120	140	160	180	200
y	0.9	1.1	1.2	1.4	1.6	1.7

Find a law of the form $y = ap + b$ using least square criteria.

Solution: To fit the straight line $y = ap + b$ by using their normal equations, let $n = 6$

$\sum y = a\sum p + bn$ (1)

$\sum py = a\sum p^2 + b\sum p$ (2)

p	**y**	**p^2**	**py**
100	0.9	10000	90
120	1.1	14400	132
140	1.2	19600	168
160	1.4	25600	224
180	1.6	32400	288
200	1.7	40000	340
900	**7.9**	**142000**	**1242**

From above equations and table, we have

$\sum y = a\sum p + bn \implies 7.9 = 900a + 6b$

$\sum py = a\sum p^2 + b\sum p \implies 1242 = 142000a + 900b$

Solving these two equations, we get, $a = 0.00814\ and\ b = 0.0952$ and put these values in given equation.

$\therefore y = 0.00814p + 0.0952.$

6. Values of x and y are tabulated as under

X	1	1.5	2.0	2.5
Y	25	56.2	100	156

Find the law of the form $x = ay^b$ to satisfy the given data by least square criteria.

Solution: To fit the parabola of the form $x = ay^b$ taking logarithms on both sides

$logx = log(ay^b)$

$logx = log(a) + log(y^b)$

$logx = loga + b\,logy$

$logx = b\,logy + log\,a$

$X = bY + c$ where $Y = logy, X = logx$ and $c = loga.$

Use their normal equations.

$\sum X = b\sum Y + cn$ and

$\sum XY = b\sum Y^2 + c\sum Y$

x	y	$X = logx$	$Y = logy$	Y^2	XY
1	25	0	1.39794	1.954236	0
1.5	56.2	0.176091	1.749736	3.061577	0.308113
2	100	0.30103	2	4	0.60206
2.5	156	0.39794	2.193125	4.809796	0.872732
7	**337.2**	**0.875061**	**7.340801**	**13.82561**	**1.782905**

From above equations and table, we have

$\sum X = b\sum Y + cn \Longrightarrow 0.87506 = 7.3408b + 4c$

$\sum XY = b\sum Y^2 + c\sum Y \Longrightarrow 1.7829 = 13.8256b + 7.3408c$

Solving these two equations, we get, $b = 0.500$ and $c = -0.6999$

But $loga = c \Rightarrow loga = -0.6999 \Rightarrow a = e^{-0.6999} = 0.49663$

Put these values in given equation.

$x = 0.49663y^{0.5}$.

7. Fit a curve $y = ax^b$ using the following data and find values of a and b.

X	2000	3000	4000	5000	6000
Y	15	15.5	16	17	18

8. If the relation between x and y is of the form $y = ab^x$. Using following values of x and y, find the values of constants a and b for the best fitting curve.

x	2.1	2.5	3.1	3.5	4.1
y	5.14	6.788	10.29	13.58	20.578

Solution: To fit the parabola of the form $y = ab^x$ taking logarithms on both sides

$logy = log(ab^x)$

$logy = log(a) + log(b^x)$

$logy = loga + x\,logb$

$logy = x\,logb + log\,a$

$Y = Bx + c$ where $Y = logy,\ B = logb$ and $c = loga.$

Use their normal equations.

$\sum Y = B\sum x + cn$ and

$\sum xY = B\sum x^2 + c\sum x$

x	y	x^2	$Y = logy$	xY
2.1	5.14	4.41	0.711	1.493
2.5	6.788	6.25	0.832	2.079
3.1	10.29	9.61	1.012	3.138
3.5	13.58	12.25	1.133	3.965

4.1	20.58	16.81	1.313	5.385
15.3	**56.38**	**49.33**	**5.001**	**16.06**

From above equations and table, we have

$\sum Y = B\sum x + cn \Longrightarrow 5.001 = 15.3B + 5c$

$\sum xY = B\sum x^2 + c\sum x \Longrightarrow 16.06 = 49.33B + 15.3c$

Solving these two equations, we get, $B = 0.3013\ and\ c = 0.07813$

But $loga = c \Rightarrow loga = 0.07813 \Rightarrow a = e^{0.07813} = 1.08126$ and $logb = B \Rightarrow logb = 0.3013 \Rightarrow b = e^{0.3013} = 1.351614$

Put these values in given equation.

$y = (1.08126)(1.351614)^x$.

9. If x and y are connected by the relation $ax^2 + by^2 = x$, find the values of a and b by rearranging the relation into linear form by using least square criteria for following data.

x	1	2	3	4	5
y	3.35	5.92	8.43	10.93	13.45

Solution: To fit the parabola of the form $ax^2 + by^2 = x$

Use their normal equations.

$a\sum x^4 + b\sum x^2y^2 = \sum x^3$ and

$a\sum x^2y^2 + b\sum y^4 = \sum x\,y^2$

x	y	x^2	x^3	x^4	y^2	y^4	xy^2	x^2y^2
1	3.35	1	1	1	11.2225	125.9445	11.2225	11.2225
2	5.92	4	8	16	35.0464	1228.25	70.0928	140.1856
3	8.43	9	27	81	71.0649	5050.22	213.1947	639.5841
4	10.93	16	64	256	119.4649	14271.86	477.8596	1911.438
5	13.45	25	125	625	180.9025	32725.71	904.5125	4522.563
15	**42.08**	**55**	**225**	**979**	**417.7012**	**53401.99**	**1676.882**	**7224.993**

From above equations and table, we have

$a\sum x^4 + b\sum x^2y^2 = \sum x^3 \Longrightarrow 979a + 7224.993b = 225$ and

$a\sum x^2y^2 + b\sum y^4 = \sum x\,y^2 \Longrightarrow 7224.993a + 53401.99b = 1676.882$

Solving these two equations, we get, $a = -1.24978\ and\ b = 0.200489$

Put these values in given equation.

$ax^2 + by^2 = x \Longrightarrow -1.24978x^2 + 0.200489y^2 = x.$